DAV Assignment-1

Name: Kimi Patle

Section: A

Batch: A1

Roll No.:07

In [ ]:
```
import pandas as pd
```

1. Data Collection .txt file and .xml file and .json file

.txt file

In [ ]:
```
with open("technology.txt","r")as file:
  content=file.read()
  print(content)
```

```
Technology has become an essential part of modern life, transforming the way
people communicate, learn, and work. The internet provides instant access to
information, while smartphones and computers help individuals stay connected
across the world. In education, digital tools support interactive learning an
d allow students to access online courses and study materials at any time. Bu
sinesses use technology to improve productivity, manage data, and provide bet
ter customer services. At the same time, responsible use of technology is imp
ortant to protect privacy and maintain cybersecurity. As technology continues
to evolve, it creates new opportunities for innovation, employment, and econo
mic growth. Learning digital skills has therefore become increasingly valuabl
e for students and professionals alike, helping them adapt to the rapidly cha
nging technological landscape.
```

In [ ]:
```
df=pd.read_csv("technology.txt",delimiter="\t")
print(df.head())
```

```
Empty DataFrame
Columns: [Technology has become an essential part of modern life, transformin
g the way people communicate, learn, and work. The internet provides instant
access to information, while smartphones and computers help individuals stay
connected across the world. In education, digital tools support interactive l
earning and allow students to access online courses and study materials at an
y time. Businesses use technology to improve productivity, manage data, and p
rovide better customer services. At the same time, responsible use of technol
ogy is important to protect privacy and maintain cybersecurity. As technology
continues to evolve, it creates new opportunities for innovation, employment,
and economic growth. Learning digital skills has therefore become increasingl
y valuable for students and professionals alike, helping them adapt to the ra
pidly changing technological landscape.]
Index: []
```

.xml file

In [ ]:
```
import xml.etree.ElementTree as ET
```

In [ ]:
```
tree=ET.parse("books.xml")
root=tree.getroot()
```

In [ ]:
```
for elem in root.iter():
  print(elem.tag,elem.text)
```

```
library

book

id 1
title Introduction to Python
author John Smith
category Programming
price 499
year 2023
book

id 2
title Web Development Basics
author Emily Johnson
category Technology
price 599
year 2022
book

id 3
title Database Management Systems
author Robert Brown
category Education
price 650
year 2021
book

id 4
title Artificial Intelligence
author Sophia Davis
category AI
price 899
year 2024
book

id 5
title Computer Networks
author Michael Wilson
category Networking
price 720
year 2020
```

.json file

In [ ]:
```
import json
```

In [ ]:
```
with open("employee.json","r")as file:
  data_json=json.load(file)
  print(data_json)
```

```
{'company': 'Tech Solutions Pvt Ltd', 'employees': [{'id': 101, 'name': 'Aara
v Sharma', 'department': 'Software Development', 'designation': 'Software Eng
ineer', 'salary': 65000, 'city': 'Nagpur'}, {'id': 102, 'name': 'Priya Verm
a', 'department': 'Human Resources', 'designation': 'HR Executive', 'salary':
45000, 'city': 'Pune'}, {'id': 103, 'name': 'Rohan Patel', 'department': 'Fin
ance', 'designation': 'Accountant', 'salary': 50000, 'city': 'Mumbai'}, {'i
d': 104, 'name': 'Sneha Gupta', 'department': 'Marketing', 'designation': 'Ma
rketing Manager', 'salary': 70000, 'city': 'Delhi'}, {'id': 105, 'name': 'Vik
ram Singh', 'department': 'IT Support', 'designation': 'System Administrato
r', 'salary': 55000, 'city': 'Hyderabad'}, {'id': 106, 'name': 'Ananya Rao',
'department': 'Data Analytics', 'designation': 'Data Analyst', 'salary': 6200
0, 'city': 'Bengaluru'}]}
```

In [ ]:
```
df=pd.DataFrame(data_json)
print(df.head())
```

```
                  company                                          employees
0  Tech Solutions Pvt Ltd  {'id': 101, 'name': 'Aarav Sharma', 'departmen...
1  Tech Solutions Pvt Ltd  {'id': 102, 'name': 'Priya Verma', 'department...
2  Tech Solutions Pvt Ltd  {'id': 103, 'name': 'Rohan Patel', 'department...
3  Tech Solutions Pvt Ltd  {'id': 104, 'name': 'Sneha Gupta', 'department...
4  Tech Solutions Pvt Ltd  {'id': 105, 'name': 'Vikram Singh', 'departmen...
```

2. Data Collection .csv file and show the content

In [ ]:

```
df=pd.read_csv("sales_data.csv")
print(df.head())
print(df.tail())
print(df.describe())
print(df.info())
```

```
   OrderID        Date   Product     Category  Quantity  UnitPrice  Total  \
0     1001  01-07-2026    Laptop  Electronics         1      55000  55000
1     1002  02-07-2026     Mouse  Electronics         2        700   1400
2     1003  03-07-2026     Chair    Furniture         1       4500   4500
3     1004  04-07-2026      Desk    Furniture         1       8200   8200
4     1005  05-07-2026  Notebook   Stationery        10         60    600

  Customer    City
0    Aarav  Nagpur
1     Siya    Pune
2   Vivaan  Mumbai
3    Anaya   Delhi
4    Kabir  Indore
    OrderID        Date     Product     Category  Quantity  UnitPrice  Total
\
10     1011  11-07-2026     Monitor  Electronics         1      12000  12000
11     1012  12-07-2026     Printer  Electronics         1       8500   8500
12     1013  13-07-2026        Lamp         Home         2        900   1800
13     1014  14-07-2026  Coffee Mug         Home         4        250   1000
14     1015  15-07-2026   USB Drive  Electronics         3        650   1950

   Customer       City
10    Arjun  Ahmedabad
11     Diya    Lucknow
12  Krishna      Surat
13    Nisha     Kanpur
14     Yash     Nashik
           OrderID   Quantity     UnitPrice         Total
count    15.000000  15.000000     15.000000     15.000000
mean   1008.000000   2.533333   6468.666667   6946.666667
std       4.472136   2.416215  13925.184721  13727.403077
min    1001.000000   1.000000     60.000000    600.000000
25%    1004.500000   1.000000    500.000000   1225.000000
50%    1008.000000   2.000000   1500.000000   1950.000000
75%    1011.500000   3.000000   6350.000000   6350.000000
max    1015.000000  10.000000  55000.000000  55000.000000
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 9 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   OrderID    15 non-null     int64
 1   Date       15 non-null     object
 2   Product    15 non-null     object
 3   Category   15 non-null     object
 4   Quantity   15 non-null     int64
 5   UnitPrice  15 non-null     int64
 6   Total      15 non-null     int64
 7   Customer   15 non-null     object
 8   City       15 non-null     object
dtypes: int64(4), object(5)
memory usage: 1.2+ KB
None
```

3. Collect dataset i.e. customer & order table, apply basic function of data preprocessing.

In [ ]:
```
# Load datasets
customer = pd.read_csv("cust.csv")
orders = pd.read_csv("order.csv")
```

In [ ]:
```
# Display datasets
print("Customer Dataset:")
print(customer)

print("\nOrder Dataset:")
print(orders)
```

```
Customer Dataset:
  CustomerID  CustomerName  Gender   Age       City             Email
0       C101  Aarav Sharma    Male  25.0     Nagpur   aarav@gmail.com
1       C102   Priya Verma  Female  28.0       Pune   priya@gmail.com
2       C103   Rohan Patel    Male  31.0     Mumbai   rohan@gmail.com
3       C104   Sneha Gupta  Female  26.0      Delhi   sneha@gmail.com
4       C105  Vikram Singh    Male  35.0  Hyderabad              NaN
5       C106    Ananya Rao  Female  29.0  Bengaluru  ananya@gmail.com
6       C107   Karan Mehta    Male  27.0    Chennai   karan@gmail.com
7       C108    Neha Joshi  Female   NaN     Jaipur    neha@gmail.com
8       C109     Raj Kumar    Male  33.0    Kolkata     raj@gmail.com
9       C110    Pooja Shah  Female  24.0  Ahmedabad   pooja@gmail.com

Order Dataset:
  OrderID CustomerID       Product     Category  Quantity  Price   OrderDate
0   O1001       C101        Laptop  Electronics         1  55000  2026-07-01
1   O1002       C102         Mouse  Electronics         2    700  2026-07-02
2   O1003       C103  Office Chair    Furniture         1   4500  2026-07-03
3   O1004       C104          Desk    Furniture         1   8200  2026-07-04
4   O1005       C105      Notebook   Stationery        10     60  2026-07-05
5   O1006       C106       Printer  Electronics         1   8500  2026-07-06
6   O1007       C107      Backpack  Accessories         2   1800  2026-07-07
7   O1008       C108      Keyboard  Electronics         1   1500  2026-07-08
8   O1009       C109       Monitor  Electronics         1  12000  2026-07-09
9   O1010       C110     USB Drive  Electronics         3    650  2026-07-10
```

In [ ]:
```
# 1. Check Missing Values
print("\nMissing Values in Customer Dataset:")
print(customer.isnull().sum())

print("\nMissing Values in Order Dataset:")
print(orders.isnull().sum())
```

```
Missing Values in Customer Dataset:
CustomerID      0
CustomerName    0
Gender          0
Age             1
City            0
Email           1
dtype: int64

Missing Values in Order Dataset:
OrderID       0
CustomerID    0
Product       0
Category      0
Quantity      0
Price         0
OrderDate     0
dtype: int64
```

In [ ]:
```
# 2. Fill Missing Values
# Fill missing Age with average age
customer["Age"].fillna(customer["Age"].mean(), inplace=True)

# Fill missing Email with 'Not Available'
customer["Email"].fillna("Not Available", inplace=True)

print("\nCustomer Dataset after filling missing values:")
print(customer)
```

```
Customer Dataset after filling missing values:
  CustomerID  CustomerName  Gender        Age       City             Email
0       C101  Aarav Sharma    Male  25.000000     Nagpur   aarav@gmail.com
1       C102   Priya Verma  Female  28.000000       Pune   priya@gmail.com
2       C103   Rohan Patel    Male  31.000000     Mumbai   rohan@gmail.com
3       C104   Sneha Gupta  Female  26.000000      Delhi   sneha@gmail.com
4       C105  Vikram Singh    Male  35.000000  Hyderabad     Not Available
5       C106    Ananya Rao  Female  29.000000  Bengaluru  ananya@gmail.com
6       C107   Karan Mehta    Male  27.000000    Chennai   karan@gmail.com
7       C108    Neha Joshi  Female  28.666667     Jaipur    neha@gmail.com
8       C109     Raj Kumar    Male  33.000000    Kolkata     raj@gmail.com
9       C110    Pooja Shah  Female  24.000000  Ahmedabad   pooja@gmail.com

/tmp/ipykernel_1390/1229909503.py:3: FutureWarning: A value is trying to be s
et on a copy of a DataFrame or Series through chained assignment using an inp
lace method.
The behavior will change in pandas 3.0. This inplace method will never work b
ecause the intermediate object on which we are setting values always behaves
as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.
method({col: value}, inplace=True)' or df[col] = df[col].method(value) instea
d, to perform the operation inplace on the original object.


  customer["Age"].fillna(customer["Age"].mean(), inplace=True)
/tmp/ipykernel_1390/1229909503.py:6: FutureWarning: A value is trying to be s
et on a copy of a DataFrame or Series through chained assignment using an inp
lace method.
The behavior will change in pandas 3.0. This inplace method will never work b
ecause the intermediate object on which we are setting values always behaves
as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.
method({col: value}, inplace=True)' or df[col] = df[col].method(value) instea
d, to perform the operation inplace on the original object.


  customer["Email"].fillna("Not Available", inplace=True)
```

In [ ]:
```
# 3. Remove Duplicate Records
customer = customer.drop_duplicates()
orders = orders.drop_duplicates()
print("\nDuplicate rows removed.")
```

```
Duplicate rows removed.
```

In [ ]:
```
# 4. Check Data Types
print("\nCustomer Data Types:")
print(customer.dtypes)

print("\nOrder Data Types:")
print(orders.dtypes)
```

```
Customer Data Types:
CustomerID       object
CustomerName     object
Gender           object
Age             float64
City             object
Email            object
dtype: object

Order Data Types:
OrderID       object
CustomerID    object
Product       object
Category      object
Quantity       int64
Price          int64
OrderDate     object
dtype: object
```

In [ ]:
```
# 5. Convert Data Types (if needed)
customer["Age"] = customer["Age"].astype(int)
orders["Quantity"] = orders["Quantity"].astype(int)
orders["Price"] = orders["Price"].astype(float)
```

In [ ]:
```
# 6. Merge Customer and Order Tables
merged_data = pd.merge(customer, orders, on="CustomerID")
print("\nMerged Dataset:")
print(merged_data)
```

```
Merged Dataset:
  CustomerID  CustomerName  Gender  Age       City              Email OrderID
\
0       C101  Aarav Sharma    Male   25     Nagpur   aarav@gmail.com   O1001
1       C102   Priya Verma  Female   28       Pune   priya@gmail.com   O1002
2       C103   Rohan Patel    Male   31     Mumbai   rohan@gmail.com   O1003
3       C104   Sneha Gupta  Female   26      Delhi   sneha@gmail.com   O1004
4       C105  Vikram Singh    Male   35  Hyderabad     Not Available   O1005
5       C106    Ananya Rao  Female   29  Bengaluru  ananya@gmail.com   O1006
6       C107   Karan Mehta    Male   27    Chennai   karan@gmail.com   O1007
7       C108    Neha Joshi  Female   28     Jaipur    neha@gmail.com   O1008
8       C109     Raj Kumar    Male   33    Kolkata     raj@gmail.com   O1009
9       C110    Pooja Shah  Female   24  Ahmedabad   pooja@gmail.com   O1010

        Product     Category  Quantity    Price   OrderDate
0        Laptop  Electronics         1  55000.0  2026-07-01
1         Mouse  Electronics         2    700.0  2026-07-02
2  Office Chair    Furniture         1   4500.0  2026-07-03
3          Desk    Furniture         1   8200.0  2026-07-04
4      Notebook   Stationery        10     60.0  2026-07-05
5       Printer  Electronics         1   8500.0  2026-07-06
6      Backpack  Accessories         2   1800.0  2026-07-07
7      Keyboard  Electronics         1   1500.0  2026-07-08
8       Monitor  Electronics         1  12000.0  2026-07-09
9     USB Drive  Electronics         3    650.0  2026-07-10
```

In [ ]:
```
# 7. Basic Statistic
print("\nSummary Statistics:")
print(merged_data.describe())
```

```
Summary Statistics:
             Age   Quantity         Price
count  10.00000  10.000000     10.000000
mean   28.60000   2.300000   9291.000000
std     3.50238   2.790858  16572.799971
min    24.00000   1.000000     60.000000
25%    26.25000   1.000000    900.000000
50%    28.00000   1.000000   3150.000000
75%    30.50000   2.000000   8425.000000
max    35.00000  10.000000  55000.000000
```

In [ ]:
```
#8. Save Cleaned Dataset
merged_data.to_csv("cleaned_customer_orders.csv", index=False)
print("\nCleaned dataset saved as 'cleaned_customer_orders.csv'")
```

```
Cleaned dataset saved as 'cleaned_customer_orders.csv'
```

In [ ]: